SOUTH LAKE HOME EDUCATORS



Lesson 11

Core Mechanics V: Wave After Wave

Semester 1, Week 11 - Level 2 Complete Milestone

At a Glance

Lesson 11 objectives & materials

By the end of today, students will:

- Nest a second decision inside an existing win-check
- Reuse spawn_enemies from a third caller: the game itself
- Keep new state (wave) on the GameManager - and reset it
- Explain what the Level 2 Complete milestone means

What We're Using Today

- Demo laptop with the four active engines verified
- Unity (C#), for the live lecture demo
- No new assets this week - reuses the spawn routine from Lesson 10
- This is a milestone lesson - Level 2 Complete!
- Your own [YourName]Game project from Lesson 10 - not a new one

IGDA framework tie-in: Programming - state machines, code reuse/refactoring, game balance and pacing (intermediate).

Today's Agenda



0:00 - 0:05	Welcome back, Lesson 10 recap (game state, the Stalemate)
0:05 - 0:20	One More Question: Is This the Last Wave?
0:20 - 0:40	The Same Spawn Code, Called a Third Way
0:40 - 0:50	Milestone Moment: What "Level 2 Complete" Actually Means
0:50 - 0:55	Introduce lab exercise, hand out engine-specific handouts
LAB TIME	***— Wave After Wave —***
0:55 - 1:50	Build multiple waves, reaching the milestone
1:50 - 2:00	Exit check: milestone celebration share-out



Welcome Back

Lesson 10 recap - your game got genuinely replayable.

5 min

Welcome Back

5 min

Quick recap of Lesson 10:

- A game_active flag that freezes gameplay, and one reset routine run by R
- Game state: everything that changed must be put back on restart
- The Stalemate: last week's Level 3 bug - still in your game, on purpose
- Today: clearing one wave becomes clearing several before you truly win

Open your own project from last week today, not a new one - and no new assets needed either, everything builds on the spawn routine you already have.



One More Question

Is this actually the last wave?

15 min

Nesting a Decision

15 min



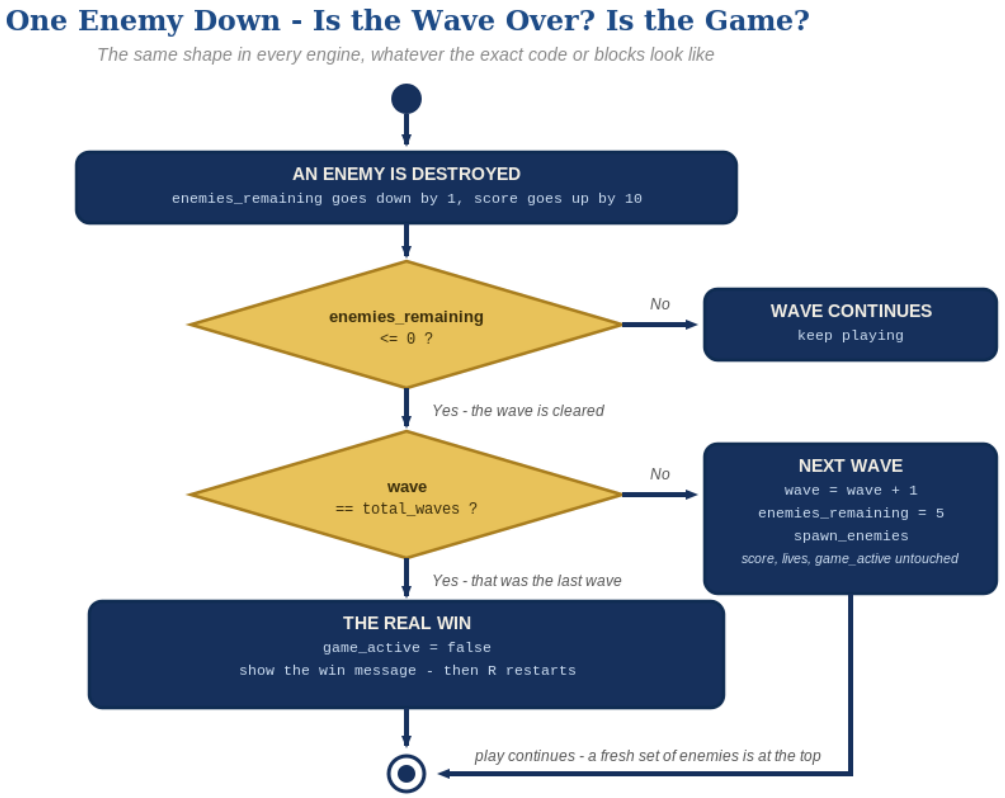
Since Lesson 6: are all enemies gone? Today, a second part: gone, AND is this the last wave? Both true -> real win. Otherwise -> the round continues.

The first time this course nests one decision inside another - a small code change, a genuinely new shape of thinking.



Discussion: can you think of a real-life decision where the answer to one question depends on a second question nested inside it? ("Can I go to the party" depends on "is my homework done," which might depend on "do I understand the assignment.")

The Whole Decision, One Picture



Two questions, in order

- Is the wave cleared? No: carry on.
- Was it the last wave? Yes: the real win.
- No: wave + 1, enemies_remaining = 5, spawn_enemies.
- Score, lives and game_active are untouched between waves.



A Third Door In

The same enemy-spawning code, one more way to call it.

20 min

Three Doors, One Room

20 min

No new enemy-spawning code today. You've had it since Lesson 5. Lesson 10 made it callable a second way, for restarts. Today: a THIRD way, every wave clear.

Unity demo: SpawnEnemies() already has two callers (Start() and ResetGame()). RegisterKill() is static, so it can't use the Spawner slot - it raises nextWaveRequested, and GameManager's Update() spawns the wave.

Same enemy-creation code. Three doors in: the very first game, every restart, and now every wave clear. That's the whole idea.

New State, Same Habit

wave and total_waves are game state too - and they live on the GameManager.

Does restart reset it?

wave must go back to 1 in the one reset routine - test it from wave 3

Who owns it?

The GameManager. Objects report; the manager counts and decides

Anything extra?

A wave message, an effect, a sound - what changes it, and what puts it back?

Same two questions

What changed when the round or wave ended? What changes it back?

Test it: reach wave 3, lose, press R. The HUD must say Wave 1.



Milestone Moment

What finishing Level 2 Complete actually means.

10 min

Level 2 Complete

10 min

Once multiple waves work today, your game can:

- Move & shoot
- Face multiple waves
- Track score
- Survive with lives
- Restart cleanly
- Truly win or lose

Level 1 MVP meant a complete, playable loop. Level 2 Complete means real depth - failure that matters, progress that builds, escalation as more waves arrive. This is the kind of thing that starts to feel like an actual arcade game, not just a working prototype.

From here through Lesson 15: sound, animation, difficulty, and polish. The mechanics are done - what's left is making them sing.



Today's Exercise & Lab

Wave After Wave - building your second milestone.

5 min

Wave After Wave

5 min

Clearing a wave spawns a fresh one, up to 3 total. Clearing wave 3 is the real win - built on top of your existing project.

Level 1 - Minimal

Clearing a wave spawns a new one, up to 3; wave shown on screen.

MVP - a complete finish for today.

Level 2 - Complete

Seamless, correct transitions every time; restart returns to wave 1.

Everything works end to end.

Level 3 - Refined

Fix the Stalemate so it works with waves; a brief wave announcement; tighter reuse.

Ready to share.



Test a FULL 3-wave playthrough today, not just wave 1 - and confirm losing a life mid-wave still works correctly no matter which wave you're on.

Lab Time Notes

- Most common bug: forgetting the wave-count check, so clearing wave 1 immediately shows the win message. Test this explicitly.
- Second most common: the wave-advance branch accidentally touching score or lives - only enemies_remaining and wave should change.
- Game-state drift: test restarting from mid-wave-2 or mid-wave-3 - restart should always return to wave 1.
- The Stalemate: confirm it is real, name it, don't hand over the fix. Level 3 is where it gets solved.
- At 1:50, do a real celebration share-out - this is the Level 2 Complete milestone. Consider 2-3 students demoing a full 3-wave playthrough.



Level 2 Complete!

Multiple waves, real stakes, a clean restart - a genuine arcade game, not just a working prototype.

Next week: sound effects - shoot, explosion, game over, wave clear. The polish begins.

Keep playtesting your full 3-wave run at home if you'd like - try a Level 3 idea if you haven't already.